Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 2 – Partial Fractions & Algebraic Methods

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Express $\dfrac{5x+7}{(x+1)(x-2)}$ in partial fractions. **[4]**

2. Express $\dfrac{3x-1}{(x-1)(x+4)}$ in partial fractions. **[4]**

3. Express $\dfrac{2x+5}{(x+2)(x-3)}$ in partial fractions. **[4]**

4. Express $\dfrac{4x-3}{(2x-1)(x+1)}$ in partial fractions. **[4]**

5. Express $\dfrac{x+8}{(x-2)(x+3)}$ in partial fractions. **[4]**

6. Express $\dfrac{3x^2+5x-1}{(x+1)(x^2+1)}$ in partial fractions. **[6]**

7. Express $\dfrac{2x^2-x+4}{(x-1)(x^2+4)}$ in partial fractions. **[6]**

8. Express $\dfrac{x^2+3x+1}{(x+2)(x^2+1)}$ in partial fractions. **[6]**

9. Express $\dfrac{4x^2 - x}{(x-2)(x^2+x+1)}$ in partial fractions. **[6]**

10. Express $\dfrac{5x^2 + 2}{(x+1)(x^2+3)}$ in partial fractions. **[6]**

11. Express $\dfrac{x^3 + 2x^2 - x + 4}{x^2 - 1}$ in the form $ax + b + \dfrac{cx+d}{x^2-1}$. **[5]**

12. Express $\dfrac{x^3 - 3x + 2}{x^2 + x}$ in the form $ax + b + \dfrac{cx+d}{x^2+x}$. **[5]**

13. Express $\dfrac{2x^3 + x^2 - 5}{x^2 - 4}$ in the form $ax + b + \dfrac{cx+d}{x^2-4}$. **[5]**

14. Express $\dfrac{x^3 + x^2 + x + 1}{x^2 + 1}$ in the form $ax + b + \dfrac{cx+d}{x^2+1}$. **[5]**

15. Express $\dfrac{3x^3 - x + 1}{x^2 - x - 2}$ in the form $ax + b + \dfrac{cx+d}{x^2-x-2}$. **[5]**

Section B – Hard (Q16–25)

16. Express $\dfrac{2x^2 - 3x + 5}{(x-1)^2(x+2)}$ in partial fractions. **[7]**

17. Express $\dfrac{3x^2 + x - 1}{(x+1)^2(x-2)}$ in partial fractions. **[7]**

18. Express $\dfrac{x^2 - 4x + 6}{(x-1)^2(x+3)}$ in partial fractions. **[7]**

19. Express $\dfrac{4x^2 - 2x + 1}{(x+2)^2(x-1)}$ in partial fractions. **[7]**

20. Express $\dfrac{5x^2 + 3x - 2}{(x-3)^2(x+1)}$ in partial fractions. **[7]**

21. Express $\dfrac{4x^3 - x^2 + 3x - 2}{(x^2+1)(x-1)}$ in partial fractions and hence find $\displaystyle\int \frac{4x^3 - x^2 + 3x - 2}{(x^2+1)(x-1)}\,\mathrm{d}x$. **[8]**

22. Express $\dfrac{2x^3 + x + 1}{(x^2+4)(x+1)}$ in partial fractions and integrate. **[8]**

23. Solve $\dfrac{2x+1}{(x-1)(x+3)} = \dfrac{1}{2}$. **[5]**

24. Solve $\dfrac{3x-2}{(x+1)(x-4)} = \dfrac{1}{3}$. **[5]**

25. Solve $\dfrac{x+5}{(x-2)(x+1)} = 2$. **[5]**

End of Worksheet • *Show all working clearly.*